

LEARNER WORKBOOK

FULL NAME & SURNAME

ID NUMBER:

NAME OF ASSESSOR

DATE OF ASSESSMENT

VENUE

ACHIEVED NOT ACHIEVED

KNOWLEDGE
SKILLS

Signature of learner

Signature of Assessor

LEARNER ACTIVITY 1

You have a monthly income of R5 500.00. Draw up a realistic monthly budget that you should be able to follow for at least eight months. The following fixed monthly expenses must appear on your budget:

- ♦ Rent - R1 500.00
- ♦ Electricity - R 185.00
- ♦ Transport - R 450.00
- ♦ Cellphone - R 375.00
- ♦ Medical aid - R 500.00
- ♦ Savings - R 400.00

Income R 5 500

Expenses

Rent R 1 500

Electricity R 185

Transport R 450

Cellphone R 375

Medical aid R 500

Savings R 400

Food R 1 000

Entertainment R 500

R 4 910 R 5 500

left with R 590.00

- Food and entertainment can change every month.
- Electricity can increase.
- Transport can increase if petrol price increase.

LEARNER ACTIVITY 2

Tom celebrated his 21st birthday in January 2008. His parents invested R10 000 in January 1997 at 9,5% p.a. interest compounded annually. They planned to use the money to buy him a car for a birthday present.

- ♦ How much money will they receive from this investment on his 21st birthday?
- ♦ How much of this money is interest?

Compound = this is when the interest earned on the money invested is added at the end of each time period to the money invested; the interest is then earned on the accumulated amount.

1997 $R10\,000 \times 9,5\%$

R 950

Total R10 950

1998 $R10\,950 \times 9,5\%$

R 1040,25

Total R 11 990,25

$$1999 \quad R 11990-25 \times 9,5\%$$
$$R 1139,07$$

$$\text{Total } R 13129-32$$

$$2000 \quad R 13129-32 \times 9,5\%$$
$$R 1247,29$$

$$\text{Total } R 14376-61$$

$$2001 = R 14376-61 \times 9,5\%$$

$$\text{Total } R 15742-39$$

$$2002 = R 15742-39 \times 9,5\%$$

$$\text{Total } R 17237,92$$

$$2003 = R 17237,92 \times 9,5\%$$

$$\text{Total } R 18875-52$$

$$2004 \quad R 18875-52 \times 9,5\%$$

$$\text{Total } R 20668$$

$$2005 - R\ 20668,69 \times 9,5\%$$

$$R\ 22632,22$$

$$2006 - R\ 22632,22 \times 9,5\%$$

$$R\ 24782,28$$

$$2007 - R\ 24782,28 \times 9,5\%$$

$$R\ 27136,60$$

Requirements achieved	YES		NO	
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Comments in support of decision

$$2008 - R\ 27136,60 \times 9,5\%$$

$$R\ 29741,58$$

On his 21 st birthday he will
receive R 29741-58

The interest will be R 19714-5

LEARNER ACTIVITY 3

Palesa considers two investment options.

- ♦ Bank A offers 7% p.a. compound interest if she invests R15 000 for five years.
- ♦ Bank B offers 8,5% p.a. simple interest if she invests R15 000 for five years.

- i) Which of these two investments earns her more interest?
- ii) How much more does she earn on the better investment?
- iii) Is compound interest always the better option? Explain your answer.

Simple interest is the amount of interest paid on the capital amount only.

<u>Bank A.</u>	<u>Bank B</u>
R 15 000	R 15 000
R 16 050	R 16 275
2) R 16 050	R 15 000
R 17 173,50	R 16 275
3) R 17 173,50	R 15 000
R 18 375,65	R 16 275
4) R 18 375,65	R 15 000
R 19 661,95	R 16 275

5) R 19661,95 R 15000
R 21038,29 R 16275

1) Bank A. Interest R 6038,29
Bank B Interest. R 6375
Bank B had more interest earned.

2) She earns R 336,71 more on bank B investment than on bank A.

3) No because the interest rate is lower than the simple interest rate.

LEARNER ACTIVITY 4

A course at university costs R10 000 rand. If a father wishes to send his daughter to do this in five years time, what will it cost him if the rate of inflation is estimated to be 11,5% p.a.?

Now	R10 000
Year 1	$R10000 \times 11,5\% = R11150$
Year 2	$R11150 \times 11,5\% = R12432$
Year 3	$R12432,25 \times 11,5\% = R13861,96$
Year 4	$R13861,96 \times 11,5\% = R15456,08$
Year 5	$R15456,08 \times 11,5\% = R17233,52$

It will cost him R17233,52
in 5 years time.

LEARNER ACTIVITY 5

Bheki's house is valued at R200 000. If houses in his neighbourhood are appreciating at a rate of 12% p.a., what can he expect to sell his house for after two years?

Now R 200 000

Year 1) $R\ 200\ 000 \times 12\% = R\ 224\ 000$

Year 2) $R\ 224\ 000 \times 12\% = R\ 250\ 880$

He can sell his house in 2 years
for R 250 880

LEARNER ACTIVITY 6

A company's machinery is valued at R950 000. If it depreciates at 5% p.a. what will it be worth after three years?

$$R\ 950\ 000 \left(1 - \frac{5}{100}\right)^3$$

$$\frac{5}{100} = 5\% \downarrow$$

$$()^3 = 3 \text{ years}$$

$$R\ 950\ 000 \left(1 - \frac{5}{100}\right)^3$$

$$R\ 950\ 000 (1 - 0,05)^3$$

$$R\ 950\ 000 (0,95)^3$$

$$0,95 \times 0,95 \times 0,95$$

$$R\ 950\ 000 (0,911525)$$

$$R\ 950\ 000 \times 0,911525$$

$$R\ 865948,75$$

The value of the machinery
will be R 865948,75 after
3 years